

Deployment Oriented Federated Learning in Connected and Autonomous Vehicles: Challenges and Opportunities

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Abstract—Connected and Autonomous Vehicles (CAVs) operate in dynamic and heterogeneous environments where sensing, computation, communication, and control must continuously adapt to changing road, weather, and traffic conditions. Traditional centralized learning faces increasing limitations in such settings, since transmitting large sensor datasets to remote servers increases bandwidth usage, raises privacy concerns, slows adaptation, and limits scalability across diverse vehicle hardware and sensor configurations. Federated learning (FL) provides a decentralized alternative that enables vehicles to collaboratively improve models without sharing raw data, supporting local training that naturally accommodates heterogeneity and facilitates continuous adaptation to rare, long-tail driving scenarios. However, deploying FL in CAVs remains challenging because training and inference must coexist during vehicle operation under strict real-time, safety, and resource constraints. This paper discusses the unique challenges of deploying FL in CAVs that arise from the tightly coupled architectural integration of CAV systems and federated learning frameworks. We outline key deployment challenges and design principles for CAV systems and FL to ensure reliable, adaptive, and safe learning in autonomous mobility.

Index Terms—Federated learning, Connected Autonomous Vehicles, Challenges, Opportunities

I. INTRODUCTION

Connected and Autonomous Vehicles (CAVs) are transforming the landscape of mobility by combining sensing, perception, planning, control, and communication into a unified framework that enables safe, efficient and collaborative mobility [1], [2]. Continuously interpreting their surroundings and making real-time decisions, CAVs operate by integrating software intelligence with vehicle dynamics, forming a closed loop from sensing to control that must meet strict real-time and safety requirements while ensuring reliability under diverse operating conditions [3], [4]. Consequently, CAV capabilities increasingly rely on data driven learning approaches that support integrated end-to-end operation [5], [6].

The restricted real-world deployment of Level 4 and 5 autonomous vehicles highlights the persistent reliability gaps in perception, prediction, planning, and control models when these models are applied across diverse operating environments, sensor configurations, and computing platforms [7], [8]. For example, Tesla Full Self-Driving (FSD) remains a supervised driver assistance feature that requires continuous driver supervision, while driverless services such as Waymo operate only within specific service areas and operating conditions [9], [10]. To support a larger deployment, these reliability

gaps must be addressed by learning from data collected across large-scale fleets under diverse operating conditions.

The current centralized CAV machine learning pipeline relies on transmitting large volumes of data to the cloud for model training. However, this approach is inefficient due to high storage overhead, limited bandwidth, and long transfer times. It also raises privacy concerns and limits scalability and adaptability in dynamic driving environments [11], [12], [13]. As autonomy level and fleet sizes continue to grow, these limitations become increasingly severe, making Federated Learning (FL) not only beneficial but also essential for future CAV ecosystems. FL enables each vehicle to train on its own locally collected data and exchange only model parameters, naturally handling distributed data diversity, including long-tail and safety-critical driving scenarios [14], [15], [16], [17], while reducing communication costs and supporting the development of privacy-preserving models [18], [19]. At fleet scale, FL sustains continual learning across diverse operating conditions and heterogeneous vehicle computing resources, improving reliability to support broader deployment [20], [21].

Although FL has been successfully deployed in privacy-sensitive applications such as healthcare and has been explored in mobile and edge systems, including CAVs [28], [29], [30], it has not witnessed its real-world deployment in CAVs. As summarized in Table I, existing FL research for CAVs as presented in surveys show largely algorithm and application centric, with limited treatment of deployment feasibility under onboard resource budgets and safety-critical timing constraints. As a result, evaluation practices commonly emphasize accuracy or convergence while paying limited attention to system metrics under heterogeneous vehicles and variable connectivity, and the coupling between inference and continuous on-vehicle training remains insufficiently examined.

The several developed FL frameworks, such as TensorFlow Federated (TFF) [31], PySyft [32], LEAF [33], Flower [34], FedML [35], FedScale [36], and NVIDIA FLARE [37], have advanced the study of FL. However, they do not account for the cyber-physical nature of CAVs, where tasks must operate under real-time and safety-critical constraints, while local training must be performed without disrupting CAV operations. Stable collaboration in CAVs requires mechanisms that can accommodate heterogeneous onboard computing capabilities, fluctuating network conditions, and mobility-driven changes in client availability. Additionally, existing frameworks provide only limited support for asynchronous

TABLE I: Comparison with Existing Surveys on Federated Learning for Connected and Autonomous Vehicles

Reference	Cross-layer heterogeneity	Resource constraints	FL coordination	Training-inference	System integration	Summary
Du <i>et al.</i> (2020) [22]	✓	✓	✓	✗	✗	Early survey of FL in vehicular IoT.
Xing <i>et al.</i> (2022) [23]	✓	✓	✓	✗	✗	Reviews the Social Internet of Vehicles, focusing on secure data sharing, trust management, and the role of FL for privacy.
Chellapandi <i>et al.</i> (2023) [24]	✗	✓	✓	✗	✗	Surveys FL in CAVs, covering data types, algorithms, and applications such as perception, trajectory, and motion control.
Zhang <i>et al.</i> (2023) [25]	✓	✓	✓	✗	✗	Surveys FL applications in ITS, covering traffic management, object recognition, service provisioning, and traffic status.
Chellapandi <i>et al.</i> (2024) [18]	✓	✗	✓	✗	✗	Surveys FL for CAVs, covering frameworks, data modalities, models, security techniques, and tasks.
Zhang <i>et al.</i> (2024) [26]	✓	✗	✓	✗	✗	Reviews FL in ITS for traffic prediction, target recognition, and edge computing, with methods, datasets, metrics.
Ali <i>et al.</i> (2025) [27]	✓	✓	✓	✗	✗	Reviews FL in CAVs, covering applications, ADAS, and security, while identifying key challenges and future directions.
OURS	✓	✓	✓	✓	✓	We present a unified CAV system-stack perspective to assess the deployment feasibility of on-vehicle FL, linking resource constraints and safety-critical real-time requirements to federated training and model updating.

Legend: ✓ = Yes; ✓ = Partially, ✗ = No

aggregation, context-aware client participation, and resource-adaptive scheduling. Thus, they do not yet enable the system-level coordination needed for the deployment of FL in CAVs.

In this positioning paper, we examine the deployment of FL in CAVs from a **distributed resource-constrained system stack** perspective. We present a unified view of the CAV software–hardware stack and analyze how resource, timing, and architectural constraints across sensing, real-time operating systems, middleware, storage, computing, and communication affect the feasibility of on-vehicle federated training and model exchange. We discuss how core FL workflow components interact with resource heterogeneity, vehicle mobility, and the coexistence of learning with real-time autonomy workloads. Finally, we identify open challenges and research directions that highlight the gap between existing FL designs and the operational requirements of real-time, safety-critical CAV systems, outlining key considerations for practical and scalable deployment.

While this paper focuses on CAVs, the insights extend to a broader class of intelligent cyber-physical systems including mobile robots, drones, and other embedded platforms, which share fleet scale distributed operation, resource constraints, and similar software–hardware architectures with real-time requirements [38], [39]. Therefore, the identified FL deployment challenges and system-level design considerations for CAVs are broadly applicable to these domains.

The rest of the paper is organized as follows. Section II presents a deployment-oriented background and motivation for FL in CAVs. Section III introduces the design principles and discusses the challenges and opportunities of integrating FL in CAVs. Section IV identifies future directions toward deployable FL in CAVs. Section V concludes the paper.

II. BACKGROUND AND MOTIVATION

The onboard software–hardware stack of a CAV operates as a tightly coupled architecture that must satisfy real-time

and safety-critical constraints [26], [40]. At the foundation of the stack, sensors, telematics, and HD maps provide continuous environment and vehicle state data, which are accessed through device drivers and managed by the operating system. Above this, the middleware layer enables standardized communication and coordination between software modules [41], [42]. These layers are executed on an onboard computing platform that provides the processing, memory, and storage resources required for real-time operation [43], [44], [45], [46]. A communication layer supports data exchange within the vehicle and between the vehicle and the external infrastructure via V2X and network interfaces [47]. The application layer implements perception, prediction, planning, and control functions to enable safe autonomous driving [48], [49], [50] as shown in Figure 1.

Despite this structured architecture, maintaining reliable performance in real-world operation introduces several system-level challenges for CAVs.

- Sensor inputs are affected by environmental variability, calibration drift, and sensor degradation, all of which impact perception reliability.
- On-board computing workloads fluctuate with scene complexity, requiring dynamic balancing across heterogeneous processors under tight power and thermal budgets.
- Real-time control depends on both stable communication and timely decision making mechanisms, as variations in latency or data delivery can directly reduce control responsiveness.
- Storage resources are limited and require selective data retention to support fleet-level model improvement while complying with memory and privacy constraints.

These challenges highlight that the CAV stack must operate under diverse and dynamic conditions, while the architecture itself remains optimized strictly for deterministic real-time inference. Therefore, additional onboard processes must operate within the constraints originally defined for safety-critical

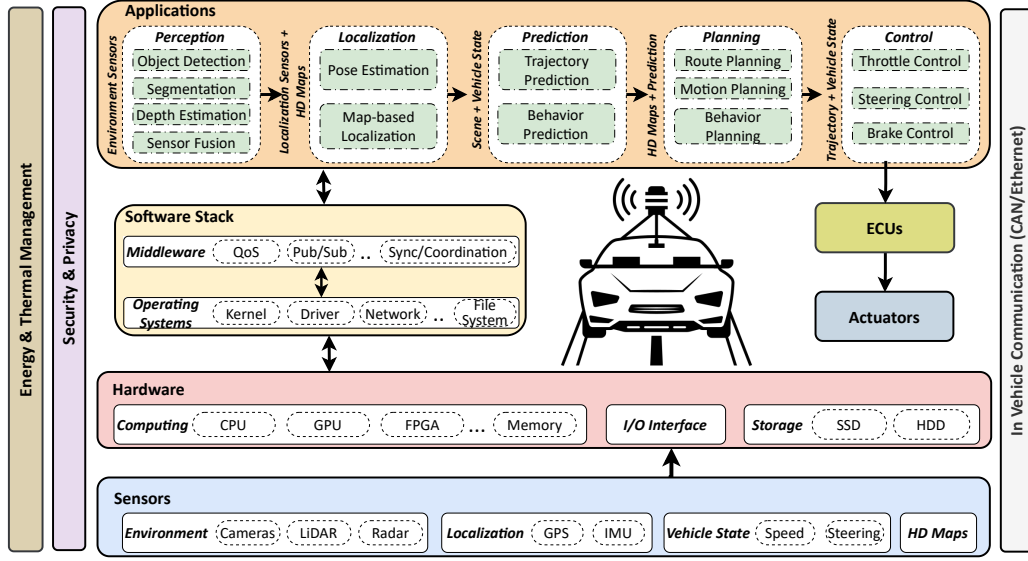


Fig. 1: Architecture of the onboard autonomous driving stack

autonomous driving functions.

Within this constrained operating environment, FL introduces a local training capability in which each vehicle refines its model using its own driving data, supporting adaptation to rare and context-specific scenarios found in the long-tail of driving conditions, while preserving privacy and accommodating distributed heterogeneity [12], [26], [51]. A global model is maintained on the aggregation server, which collects local parameter updates from vehicles, aggregates them, and distributes the updated parameters back to the fleet [18], [20], [52], [53], as shown in Fig. 2. The variability in connectivity and resource availability across the fleet requires asynchronous aggregation so that updates can be incorporated without requiring synchronized participation. In large-scale deployments, exchanging updates between many vehicles increases the communication load. Hierarchical aggregation addresses this by having roadside units (RSUs) aggregate local updates from nearby vehicles before sending the aggregated results to the global server [54], [55], [56]. Real-time inference operates under strict timing and safety-critical constraints, local training must be scheduled to ensure that these constraints are met.

Federated training in CAVs is primarily constrained by system-level factors.

- Federated training must be coordinated with real-time inference to maintain latency for safety-critical workloads.
- Vehicles exhibit computing heterogeneity, which restricts the scale and frequency of local training.
- Data across vehicles are non-IID, as different driving environments produce different local model updates.
- Exchanging model updates introduces communication overhead under limited and intermittent connectivity.
- Training must remain robust to adversarial or faulty clients while preserving privacy.

The practicality of FL is limited by the system’s ability to maintain learning under variable computing capacity, communication, and environmental conditions.

Enabling FL in CAVs requires system-level coordination

rather than algorithmic improvements alone. Local training and model exchange must be integrated into the autonomy stack and preserve real-time and safety requirements, while also adapting to heterogeneous resources and intermittent participation across the fleet. Vehicle mobility and evolving scene complexity create highly variable connectivity and shifting real-time autonomy workloads. This variability competes for onboard resources and thereby limiting when local training can execute and when secure model updates can be applied or exchanged. These system conditions determine when vehicles can participate in training and how updates can be incorporated safely. Thus, FL in CAVs becomes a **system integration challenge** in which learning must coexist with time-critical autonomy workloads under heterogeneous and dynamic operating conditions.

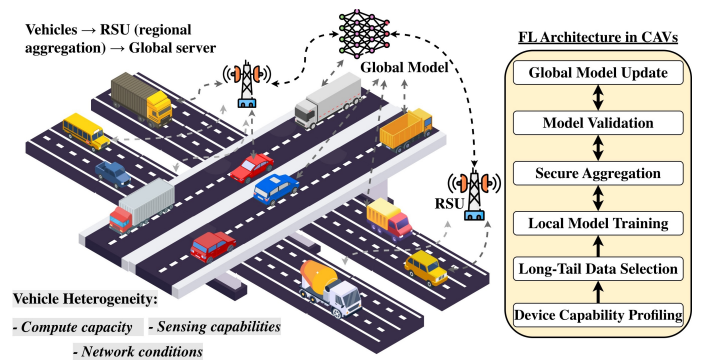


Fig. 2: Federated learning in connected autonomous vehicles

As summarized in Table I, existing FL research has primarily focused on algorithmic design, aggregation strategies, and privacy mechanisms, while giving limited attention to system-level integration required to support learning alongside real-time autonomous driving workloads. Current frameworks do not coordinate learning with on-board computing, communication, and safety constraints in operational CAV settings. This motivates a deployment-oriented perspective in which learning

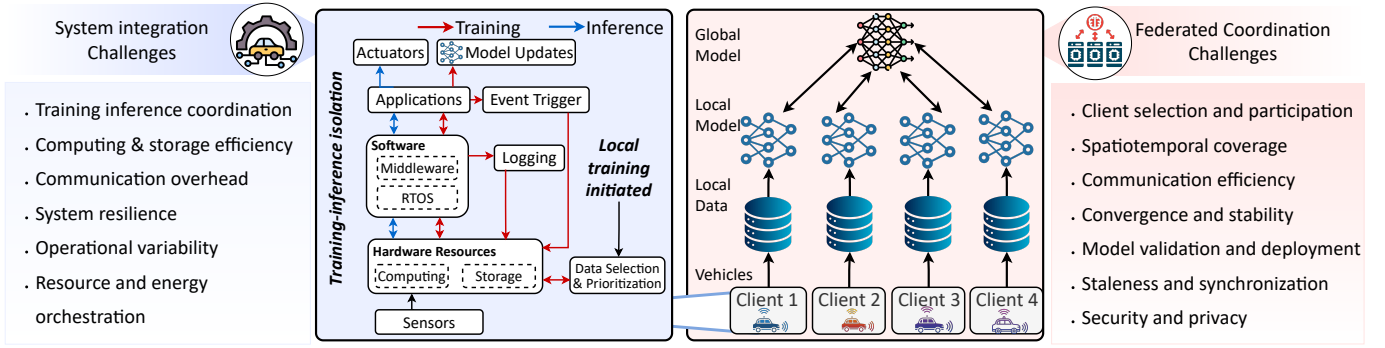


Fig. 3: System level architecture and coordination workflows for FL in CAVs and design challenges

is co-designed with runtime operation, resource allocation, and safety guarantees to ensure reliable and scalable autonomous driving.

III. CHALLENGES AND OPPORTUNITIES

The effective integration of federated training into autonomous vehicle requires careful consideration of how training can occur alongside safety-critical inference as shown in Figure 3. The training pipeline initiates with the collection of raw sensor streams, after which selective logging and data selection determine which samples are used for updating the local model. The inference pipeline must operate continuously and comply with strict latency constraints, whereas training scheduled during operational modes that reduce risk and resource contention. These include idle or parking states, periods of low computational demand such as highway cruising, or conditions where dynamic resource scaling temporarily increases available computing power or energy. In such contexts, training can proceed without compromising vehicle performance or safety.

The deployment of FL in CAVs introduces several architectural, data, and resource challenges. The following principles outline the key requirements for this integration and provide a foundation for the challenges that follow.

- Driving behavior must remain stable, predictable, and safety consistent while FL updates the onboard models, and model adaptation must never interfere with inference.
- FL in CAVs must operate with full awareness of system heterogeneity and operational constraints, adapting to dynamic resource conditions and connectivity to ensure that learning remains feasible and aligned with vehicle runtime environment.
- FL in CAVs must accommodate capability differences across the fleet, allowing each vehicle to participate at a level appropriate to its hardware and resource budget, while avoiding overload on weaker platforms and over reliance on strong ones.
- Client participation and data selection should follow the operational variability, scenario relevance, and safety-critical task requirements, while ensuring that infrequent critical events are retained in training despite the dominance of common scenarios.
- FL must preserve system-level integrity under continuous adaptation, ensuring that updates remain consistent

with vehicle operational constraints and fleet long-term reliability requirements.

Based on these principles, challenges and opportunities are discussed in relation to the structure of the CAV stack and the FL workflow. The autonomous driving stack comprises the real-time operating system, middleware, storage, onboard computing, and application level datasets and models, and this layered organization is critical because FL must operate within the real-time, resource, and safety constraints defined at each level. In addition, FL introduces data and client selection and global aggregation across vehicles, while communication, security, and privacy span both the CAV stack and FL, affecting on-vehicle data movement, external connectivity, and the protection and validation of shared model updates.

A. FL Challenges Induced by CAVs Architecture

In this subsection, we examine the challenges from the CAV system stack, emphasizing the component level constraints that shape local training and inference. As shown in Table II, we selected three representative CAV system stacks that reflect commonly used configurations in current practice. However, these platforms were not originally designed with federated coordination or on-vehicle collaborative model training. As a result, enabling FL requires rethinking how computation, data movement, and communication interact within the vehicle. To support FL operation in CAVs, the system stack must be carefully coordinated. We therefore begin by examining key system-level components, real-time operating systems, middleware, storage, hardware resources, and applications.

1) Real-Time Operating Systems

Real-Time Operating Systems (RTOS) serve as the foundation for resource coordination and execution control in CAVs, mediating interactions between hardware, real-time applications, and machine learning workloads. QNX and VxWorks are widely adopted in the automotive industry [44], [57]. QNX [58] uses a microkernel architecture in which essential functions such as CPU scheduling, interprocess communication, and interrupt management are isolated from user level processes. VxWorks provides predictable timing behavior, supports multiple processor architectures, including Intel, Power, and ARM, and meets safety and security certification requirements for embedded automotive systems [59]. NVIDIA Drive OS combines an embedded real-time operating system, hypervisor, CUDA, and TensorRT libraries

to accelerate machine learning workloads for perception and control in autonomous vehicles [44], [60]. However, current real-time operating systems are not fully equipped to support federated learning workloads, as they lack native capabilities for **distributed coordination, secure communication, and adaptive resource management** across heterogeneous edge nodes. Therefore, operating systems must evolve to enable seamless integration of FL in CAVs.

At the system-level, **scheduling and isolation** pose critical challenges for **training-inference coordination** when integrating federated training into real-time CAVs. Inference must operate as a hard real-time process with strict timing guarantees, while local training functions as a soft real-time or best effort workload. This requires scheduling policies that maintain predictable inference latency under variable training loads. Existing priority and partition based schedulers ensure determinism for safety-critical tasks but remain tailored to static, intra-vehicle task graphs on homogeneous control units [63], [64], [65]. FL for CAVs introduces highly dynamic compute demands, where training and inference compete for limited computational resources. Managing this sharing is a core scheduling challenge for RTOS, requiring adaptive resource-aware policies that preserve inference deadlines while utilizing idle compute periods for training.

Federated workloads introduce **execution level variability** that challenges kernel and driver subsystems in real-time environments. Dynamic memory allocation, asynchronous compute calls, and frequent CPU accelerator data transfers create irregular access patterns that conventional kernels are not equipped to handle. Studies on accelerator based real-time systems [66] show that these transfers are typically non-preemptive, preventing the OS from pausing long data movements to service higher priority inference tasks. Such constraints extend to the driver layer, where components tuned for steady inference workloads face contention and I/O blocking during local training. To maintain deterministic timing, the RTOS must implement predictive I/O control, bandwidth isolation, and asynchronous scheduling to coordinate resource access across concurrent inference and training processes.

Memory and intra-system synchronization are essential RTOS functions for maintaining real-time predictability. The RTOS must bound allocation latency, isolate memory used by safety-critical inference tasks, and regulate concurrent access to shared resources through priority-aware synchronization. These mechanisms prevent interference between inference and training processes, ensuring deterministic behavior under mixed workloads.

System resilience is an essential requirement for RTOS to manage the dynamic behavior of FL workloads. Failures during local training must be contained and recovered without disrupting real-time inference or control loops, ensuring system stability and responsiveness. Developing RTOS with native support for fault isolation, recovery, and adaptive resource control remains a challenge for enabling safe and scalable FL in CAVs.

2) *Middleware*

The middleware forms the backbone of the CAV software stack by managing communication between sensors, compu-

tational modules, and control systems. The Robot Operating System (ROS) and ROS 2 provide flexible open source middleware that abstracts hardware complexity and enables modular data exchange through a publisher, subscriber model [67], [68], [69]. However, because ROS lacks deterministic timing and safety compliance, domain and industry specific frameworks have emerged. Autoware extends ROS into a complete autonomous driving stack, integrating perception, localization, planning, and control with standardized interfaces and real-time support [70]. In contrast, Apollo Cyber RT and AUTOSAR Adaptive target production grade autonomy by offering deterministic, safety certified, and service oriented communication [41]. Together, these frameworks bridge low-level hardware and high-level decision layers, enabling scalable and reliable autonomous vehicle operation.

Real time autonomy software is typically run under a timing contract with fixed periodic tasks and **bounded end to end processing time**, and any overrun is treated as jitter or a missed deadline that can require mitigation [42], [71]. For example, Betz *et al.* measure the end-to-end runtime latency in the Autoware ROS 2 and report 152 ms mean, 322 ms p99, and 402 ms max, with idle latency comprising 57.8% of the budget and reflecting inter module waiting [72]. The measured middleware latency budget indicates limited timing margin, making it essential to tightly control the additional computation and communication overhead introduced by federated learning.

As CAVs evolve toward FL, middleware must extend beyond deterministic inference support to accommodate on-vehicle training and model exchange. This transition introduces significant **computational and communication overhead**, as the same serialization pipelines that currently transmit lightweight perception data must now handle large gradient tensors, feature maps, and model checkpoints without violating real-time constraints. Efficient **bandwidth allocation, serialization efficiency, and QoS-aware message routing** therefore become design challenges to deploy FL in CAVs.

A fundamental design principle to enable FL in CAVs, middleware is functional **isolation between inference and training layers**. The inference layer must remain deterministic and real-time to guarantee safety-critical responsiveness in perception and control, whereas the learning layer can tolerate update **staleness and synchronization** delays during training. Achieving this separation requires middleware capable of managing concurrent data and model pipelines for sensing, training, and actuation while employing lightweight serialization mechanisms optimized for high frequency data exchange. Dynamic policy scheduling and adaptive bandwidth management are further required to balance computational and FL demands so that model updates do not interfere with inference performance.

To enable FL within CAVs, middleware must evolve into a learning aware communication foundation, one that can prioritize safety-critical inference while opportunistically executing training and model synchronization under resource constraints. Such an architecture would transform middleware from a passive data transport layer into an active, intelligent coordination layer, capable of maintaining real-time autonomy

TABLE II: Automotive System on Chip (SoC) Platforms and System Stack Components

Class	Platform	RTOS	Middleware	Compute Architecture
Automotive SoCs (Commercial) [61]	NVIDIA DRIVE (SoC - Orin, Thor)	Linux (DriveOS) + QNX RTOS for safety	DriveWorks	CPU, GPU, DLA (Deep Learning Accelerator), PVA (Programmable Vision Accelerator), OFA (Optical Flow Accelerator)
Automaker Designed (OEM) Automotive SoCs	Tesla FSD HW3 / HW4	Tesla proprietary RTOS	Tesla proprietary Middleware	Tesla FSD SoC (Full Self-Driving System on Chip)
Legacy Development Platforms (R&D) [62]	NVIDIA Drive PX2 Pegasus	Linux + QNX	Early DriveWorks	CPU, GPU, DLA, PVA, VIC (Video/Image Processor)

while supporting continuous learning on the edge.

3) Storage

In FL for CAVs, storage supports both hard real-time inference and soft real-time local training. Inference requires timely access to the data and model, while local training requires longer time to complete and may delay due to inference priority or system constraints. When training is delayed, incoming sensor data and intermediate updates must remain in storage for longer periods, increasing storage residence time. Moreover, data collection rates vary significantly across vehicles due to differences in routes, environments, and driving conditions, resulting in unequal storage demands. These challenges are amplified by **system heterogeneity** across vehicles in storage and compute capacity across vehicles, which leads to non-uniform data retention requirements and local training progress.

FL in CAVs adopts a three layer storage hierarchy consisting of cache, on-board solid state storage (SSD), and an extended storage layer (HDD) [73]. The cache stores frequently accessed model parameters and intermediate feature activations required to sustain deterministic inference latency. The SSD layer maintains the subset of recently collected sensor data that is prioritized for near term on-vehicle training, while the HDD layer retains longer term data archives to support federated training and the preservation of rare or safety-critical events that are needed for future model refinement.

Wang et al. [74] quantify storage demands from real sensor and CAN traces and derive a multi-layer storage architecture, reporting aggregate multimodal data rates of 14.424 TB/day (i.e., 166.915 MB/s); even with selective retention for training, these streams still impose significant challenges for storage, access, and processing. On NVIDIA Jetson AGX Orin, FLEdge [75] reports per round client computation time of 11s for FEMNIST (CNN) and 324s for SAMSum (FLAN-T5 Small), indicating that local training can consume tens to hundreds of seconds per round and thus increase storage residence time for staged data and intermediate artifacts.

TABLE III: Storage Hierarchy

Real-Time	Layer [73]	Capacity	Latency
Hard [73]	Cache	16GB	Fastest
Soft [76], [77]	SSD (UFS Flash)	256 GB	Moderate
Non-Real Time	HDD	TBs at scale	Slow

We use NVIDIA DRIVE AGX Orin as a representative platform, as summarized in Table III, due to its publicly documented storage and memory configuration [76], [77]. Under high sensing rates and time extended local updates, cache amplifies contention because inference and training compete for a small, fastest shared working space, while SSD **amplifies**

residence time because selected data can accumulate faster than it is processed.

Federated training relies on the data that is selectively collected onboard, particularly data from long-tail, rare events, and safety-critical scenarios. Vehicle computing resources may restrict the volume of data that can be used in each training cycle. Therefore, optimal **data placement across storage layers** remains an open challenge, requiring adaptive policies that balance workload variability, vehicle heterogeneity, real-time constraints, and storage endurance in federated training.

As data volumes grow, compression is necessary to manage storage across the SSD and HDD layers; however, it must be carefully designed, as perception models rely on fine spatial and temporal features, so that compression does not compromise accuracy. **Task-aware compression** is therefore required to lower storage footprint while preserving information essential for effective training. Even with compression, long term retention remains challenging, as data can accumulate faster than can be used for training. Retaining all data indefinitely is infeasible, but discarding data too early risks losing rare or safety-critical samples that are valuable for future model updates. Therefore, **retention** should follow adaptive lifecycle policies that prioritize, compress further, or discard data based on relevance and storage pressure.

To make the storage hierarchy practical for federated learning in CAVs, storage management must be integrated and treated as a core element of the FL pipeline. In practice, this involves coordinating data selection, compression, retention, and training scheduling in response to real-time inference load, available compute resources, and changing value of stored data. Aligning these decisions with safety-critical inference requirements and training opportunities ensures that high value data remain accessible while the system continues to meet real-time performance constraints.

4) Computing

CAVs rely on onboard hardware to perform perception, prediction, planning, and control tasks that require high computational capability and strict timing assurance. Each vehicle hosts a diverse set of computing resources (CPUs, GPUs, FPGAs, and specialized processors) to execute complex workloads [78], [79]. However, modern automotive platforms increasingly integrate these heterogeneous computing units into system-on-chips (e.g., NVIDIA DRIVE Orin, Qualcomm Snapdragon Ride), combining CPUs, GPUs, and AI accelerators on a single chip to improve efficiency and reduce communication overhead [44], [80], as shown in Table II. Although modern hardware offers high performance per watt, its cost and integration complexity hinder large scale deployment in practice. Thus, while hardware capability has advanced con-

siderably, the balance of affordability, scalability, and energy efficiency remains an open challenge for the adoption of FL in CAVs.

As reported in iFLOW’s multi-model FL setting [81], profiling indicates that on-vehicle training demand varies across clients and across epochs. For three edge clients, Client 1 exhibited high GPU utilization at the beginning of training (up to 90.5%) and subsequently operated in the 11.5%–17.1% range, with CPU usage 4%–35%, Client 2 maintained 85% memory utilization with GPU 11.6%–18.2%, and Client 3 maintained 61%–63% memory utilization with GPU 12%–18%. Epoch duration ranged from 20–78s. Prior profiling studies show that resource demand depends strongly on workload configuration, affecting CPU, GPU, memory usage, training time, and accuracy [20].

These profiling observations highlight that on-vehicle FL introduces non-uniform compute and memory pressure across clients and epochs. Consequently, **system heterogeneity** creates challenges in maintaining consistent performance for FL, where both **inference and training compete for shared resources**. Balancing the hard real-time requirements of inference with the adaptable nature of local training while ensuring safe and efficient resource utilization remains a fundamental research challenge in enabling scalable, learning enabled CAV deployments. Training requires additional computing and memory resources to support both inference and training without compromising real-time performance. Inference and training often differ in how onboard computing units are utilized and prioritized, even when executed on shared hardware, requiring careful **system-level coordination** of computational resources. In addition, the required level of memory and accelerator capacity remains an open research problem because CAVs perform three core functions, perception, decision making, and control, which differ in computational demands, and these workloads vary with changing sensor configurations, driving conditions, and model updates.

Scaling computing and memory resources to support both inference and training inevitably increases system complexity, power demand, and thermal load, all of which must be managed within the **operational variability and energy limits** of the vehicle. Sustained high power workloads can accelerate battery degradation and reduce driving range, highlighting the need for efficient power allocation and temperature control. Supplementary energy systems such as solar assisted charging [82] can provide support but remain dependent on design and environmental conditions. Balancing hardware capacity, task scheduling, and energy management remains essential for scalable and sustainable FL in CAVs. However, a unified framework that coordinates resource allocation, multi-module training, and energy constraints under real-time requirements is still underexplored and remains a key research challenge for scalable learning enabled CAV systems.

5) Applications

At the system-level, FL serves as a continuous fleet wide adaptation mechanism that complements the real-time inference stack in CAVs. The CAV autonomy tasks include perception, prediction, planning, and control, each relying on distinct data streams, learning objectives, and computational charac-

teristics, while contributing to overall system performance. A critical challenge is determining how to schedule training across these tasks, since they depend on **heterogeneous data modalities** and streams, have different latency and resource requirements, and jointly affect overall system performance. Training all modules simultaneously can overwhelm onboard resources, while training them individually risks incomplete adaptation. Balancing these training demands remains an open problem. This uncertainty has direct implications for hardware scalability and architectural design.

For perception, FL enables vehicles to improve object detection, segmentation, and scene understanding models by training on diverse environmental conditions observed across the fleet. This collaborative learning is particularly valuable for long-tail and safety-critical scenarios, such as adverse weather, night time driving, glare, or occlusion that occur infrequently for any single vehicle, but are collectively encountered across the fleet [83], [84], [85], [86]. As a result, these perception models can better maintain detection accuracy and scene interpretation consistency across variable and dynamic conditions present in real-world driving.

For prediction, planning, and control tasks, FL allows vehicles to refine prediction, trajectory planning, and motion control models using diverse driving experiences gathered across the fleet. This shared learning improves the reliability of behavior prediction, maneuver selection, and vehicle response in varying traffic conditions, environments, and vehicle states [87], [88]. However, the effectiveness of these applications is fundamentally determined by the quality and diversity of the data available for training and the design of the models used to learn from it.

B. FL Framework Induced Challenges in CAV Systems

We now focus on the challenges of the federated learning framework in CAVs. FL in CAVs must operate under a unified resource budget that includes sensing, preprocessing, computing, energy, and thermal limits, ensuring that distributed training and communication remain feasible without compromising real-time autonomy or system safety [18], [27].

CAVs generate data on a petabyte scale per year, only a tiny fraction well below one percent is ultimately used for training after preprocessing, yet this reduced data volume requires computation to enable model training [89]. However, real-time inference is safety-critical and must always retain priority. To preserve this requirement, data and metadata selection for training is triggered only after inference execution, while computational resources are concurrently shared under controlled coordination. This safety first design enables inference and training to coexist without compromising the reliability or responsiveness of the inference pipeline.

CAVs are inherently heterogeneous, making federated learning coordination a complex and dynamic process [51], [90], [91]. Addressing this heterogeneity requires workflow policies for data selection, client selection and participation, aggregation, and model management that ensure stable learning under non-IID data, maintain **spatiotemporal coverage**, and enable **safe validation and deployment**, while accounting for

secure aggregation and cross module consistency. We therefore structure the following discussion along the FL workflow in CAVs: datasets, models, data selection, client selection, aggregation, and global model management.

1) *Datasets*

Dataset design and evaluation protocols remain central challenges in practical FL for CAVs. Large scale autonomous driving datasets such as KITTI [92], BDD100K [93], nuScenes [94], Argoverse [95], Waymo Open [96], A2D2 [97], and ApolloScape [98] have advanced autonomous driving research, but they are largely curated for centralized training. As a result, they often fail to reflect federated deployments with decentralized, heterogeneous clients, non-IID data, and evolving temporal conditions. Moreover, these benchmarks under represent rare and safety-critical long-tail events, limiting the ability to evaluate FL strategies to discover, prioritize, and learn from edge case scenarios across fleets. These challenges motivate the development of datasets and evaluation protocols that explicitly represent **client heterogeneity, multimodal variability, temporal drift, and long-tail event prevalence** in real-world CAV environments.

The data annotation process represents another significant limitation. Generating high quality labels for driving scenes requires significant human effort and domain expertise, and this challenge is further intensified in decentralized settings where each vehicle demands localized annotations. Therefore, semi-supervised [99], [100], self-supervised [101], [102], [103], and unsupervised [104], [105] methods become essential to reduce manual labeling while maintaining model reliability. Incorporating these approaches within FL enables vehicles to collaboratively learn from partially labeled or unlabeled data while preserving data privacy. However, integrating semi or self-supervised learning with FL is non-trivial, as **heterogeneous driving environments and inconsistent annotation policies** can lead to feature and label noise across clients. In addition, noisy auto-generated labels can then be amplified over federated training rounds, highlighting representation alignment and reliability aware aggregation as a key challenge.

2) *Models*

From a modeling perspective, most existing architectures are designed for homogeneous computational environments and centralized training pipelines. In federated CAV systems, hardware diversity extends from high performance edge servers with GPUs to embedded units with limited processing and memory capacity. Model training requires considerably higher computational and energy resources than inference, posing a major challenge for deployment in **resource-constrained** vehicles. Since most vehicles are optimized for inference, achieving efficient local training demands hardware aware optimization and adaptive model design. Federated CAV models should therefore emphasize lightweight and resource-efficient architectures that maintain strong representational capacity under real-time and energy constraints.

Scalability and client adaptation of these models are central challenges in federated deployment. A model with high computational cost may perform well on powerful clients yet be infeasible for resource-constrained vehicles. The parameter or resolution reduction can reduce cost, but often reduces

detection accuracy [20], which motivates FL frameworks that preserve the trade off between accuracy and efficiency on heterogeneous hardware. Federated CAV systems can take advantage of model compression and transfer techniques to meet on-vehicle computing and energy budget, such as pruning [106], knowledge distillation [107] to reduce complexity while retaining task performance. However, pruning and distillation targets can vary across vehicles, creating **inconsistent local updates and weakening aggregation**, which motivates the development of aggregation compatible compression and distillation strategies for federated CAV training.

In **dynamic driving environments**, continuous operation motivates continuous and lifelong learning, yet models trained on sequential or evolving data can overwrite previously acquired representations, leading to catastrophic forgetting. This challenge highlights the need for architectures capable of retaining essential knowledge while adapting to new information. Incorporating continual and lifelong learning strategies within model design can enhance stability and preserve long-term performance across dynamic environments. They should also remain resilient under diverse driving conditions such as long-tail object distributions, motion blur, occlusion, and adverse weather through multimodal and cross-domain learning strategies that integrate information from cameras, LiDAR, and radar.

Convergence behavior remains a fundamental challenge under **non-IID data, heterogeneous computing capacity, and asynchronous or stale updates**. Deep architectures trained on evolving and heterogeneous data often exhibit unstable optimization, slower convergence, and sensitivity to learning-rate variations. These challenges are intensified by the diverse sensor inputs, dynamic driving conditions, and resource disparities across vehicular platforms. Improving **convergence stability** requires model architectures that are adaptable to data diversity and computational variability, enabling consistent and efficient learning across dynamic and resource-constrained environments.

3) *Data Selection and Prioritization*

Data selection is a vehicle level problem that plays a critical role in determining the efficiency and robustness of FL in CAVs. The learning process should emphasize safety-critical and underrepresented long-tail scenarios events that occur infrequently during normal driving but disproportionately influence generalization and failure modes, such as low-light conditions [108], glare [85], occlusion, and other challenging perception situations that represent rare and important driving contexts. Previous work has explored data and feature reduction primarily from the perspective of improving model training efficiency [109], [110], but lacks scenario awareness. While long-tail events should be prioritized for training, identifying such events itself requires computation. Therefore, long-tail data selection is a key challenge because it must balance scenario relevance, computational cost, and real-time system constraints. Given the disproportionate impact of rare events on generalization, this becomes a primary motivation to address the problem.

In addition, the selection of frames should account for vehicle dynamics, where the sampling frequency can be

adapted on the basis of speed to capture informative visual variations while avoiding unnecessary data accumulation for federated training. Since perception, decision-making, and control modules are closely coupled, it remains an open question whether shared training on the same scenarios yields better coordination or whether task specific data subsets are sufficient. Exploring this relationship is essential for enabling coherent and **resource-aware learning** across the autonomous driving stack.

Data selection must be complemented by data prioritization. Under **system heterogeneity, mobility-driven workload variability and communication constraints**, vehicles must allocate limited onboard resources between autonomy functions and learning tasks to support timely local training and update transmission [111], [112]. Therefore, even within selected long-tail data, prioritization is necessary to schedule the most safety-critical events (e.g., near conflict or sudden braking) ahead of less urgent or redundant rare scenarios to improving learning efficiency while respecting constraints. Prioritizing within selected long-tail data remains an open research problem, long-tail scenarios are highly diverse, and their relative importance varies across operating contexts and heterogeneous vehicles, requiring system-level mechanisms to prioritize rare events under constrained and intermittent participation.

4) Client Selection

The selection of clients determines the vehicles that participate in each federated training round and directly influences the quality, diversity, and stability of the global model. Since each vehicle differs in onboard computing capability, energy availability, communication bandwidth, and sensing configuration, client selection must account for this heterogeneity. Existing work explores client selection as resource and mobility aware [113] with data quality by prioritizing vehicles with high image quality to meet vehicular edge constraints [114] or reliability-aware [115]. However, even advanced vehicles with richer sensors and higher processing capacity may not capture the complete range of environmental variations. Client selection therefore must account for broader **fleet coverage and event-aware participation**. For example, a low capacity vehicle with limited sensing capability may record a rare or safety-critical event that a high end vehicle passing earlier on the same route does not observe. These circumstances raise two key challenges:

- Recognition of missed events that require retrospective inclusion into the learning process while maintaining consistency and convergence reliability.
- The reliance on vehicles with greater computational or sensing capability must be guided by **fairness aware participation policies** to ensure that their contributions enhance global model quality without introducing systemic imbalance

Mobility aware participation and resource-aware coordination have been explored in recent vehicular FL frameworks, which account for intermittent connectivity (dropout and rejoining) and heterogeneous onboard compute and network conditions [81], [116]. However, scaling these mechanisms to fleet deployment remains nontrivial. In particular, rejoining

should be explicitly designed to remain robust when vehicles return after long absences with stale local states, while still prioritizing updates derived from rare, long-tail, or safety-critical events that may be underrepresented in the fleet. The connectivity varies geographically (e.g., cities, rural), so many vehicles in the same area can drop out simultaneously. An open research problem is to design client selection that remains robust to such **region wide dropouts** by mitigating connectivity driven participation and enabling stable reintegration when the region reconnects.

Addressing these issues is essential for developing a globally consistent and event aware federated model that unifies information across vehicles with differing capabilities. Client participation must be aligned with **resource availability, intermittent connectivity** and update timing to avoid aggregation delays and inconsistent model states during vehicle operation.

5) Aggregation

Federated aggregation enables fleet level learning by integrating vehicle updates into a continually improving global model. In CAV deployments, mobility and uneven on-vehicle computing capacity lead to intermittent connectivity and highly variable update latencies, making strict synchronization impractical; therefore, aggregation relies on asynchronous coordination [117], [118], [119]. **Asynchronous aggregation** must be sensitive to **staleness**, and updates computed on older global models should be downweighted, discarded beyond a delay limit, or smoothed to prevent drift and oscillations.

Under these conditions, the aggregation should be explicitly aware of resource and heterogeneity, incorporating differences in onboard capacity to preserve **training stability**, avoid disrupting real-time inference, and maintain efficient learning. Under asynchronous coordination, aggregation should also avoid disproportionately weighting high capacity vehicles that contribute more frequently to support consistent global updates. In addition, the frequency of aggregation updates must balance responsiveness with stability, since overly frequent updates can trigger **oscillatory behavior** and increase communication load, whereas infrequent synchronization yields **stale global models** and slower alignment. Accordingly, aggregation must be engineered as a resource-aware coordination layer that delivers stable global convergence while remaining compatible with real-time inference and safety constraints.

The updates arise from diverse driving conditions, including weather, traffic, and lighting, and a single global aggregation rule may fail to capture condition specific behaviors. **Condition aware hierarchical aggregation** maintains context specific sub-models by clustering vehicle updates based on operating conditions. Each cluster is aggregated separately, and the resulting models are aligned with their corresponding environments. These considerations motivate hierarchical and condition aware aggregation as core design principles for scalable federation and environment aligned model learning in CAVs.

6) Global Model

The deployment of global models is a critical control point in federated autonomous driving. As large-scale deployment expands, **fleet level management** becomes increasingly complex, requiring mechanisms to track active and inactive

participants, diagnose non-participation causes, and maintain coordination consistency across dynamic vehicle networks.

The global update of the model remains challenging due to heterogeneous participation, and asynchronous delayed updates can introduce **staleness and instability** in the aggregated model [120]. **Convergence** depends on how much training is performed locally versus how frequently updates are exchanged: increasing local training reduces communication frequency but raises on-vehicle computing cost and can increase drift, whereas more frequent synchronization improves update freshness but increases bandwidth usage and aggregation overhead at scale. This motivates the development of FL policies that balance update **freshness, convergence stability, and resource constraints under dynamic vehicular connectivity**. However, before any global model is redistributed to participating vehicles, its performance should be verified against curated and labeled datasets to ensure functional correctness and mitigate safety-critical risks or adversarial behavior.

The updated parameters of the global model should be deployed to vehicles in a timely and controlled manner [121], [122], to ensure that the inference modules operate on a current validated model version, with consistent update timing and minimal communication latency throughout the network. Release timing is a key system decision at scale, since an updated global model may not be immediately suitable for deployment under heterogeneous connectivity and bandwidth constraints, practical systems require explicit policies that determine when a model is eligible for release based on measured gains, stability criteria, and network conditions. Global model updates should be **versioned and deployed via staged rollouts**, with rollback to a known safe prior version when post deployment monitoring detects regressions, instability, or safety violations. **Global model deployment** should be managed as a safety-critical process with explicit validation, policy-driven release timing, controlled rollout, and rollback.

These mechanisms establish the foundation for an adaptive and resource efficient FL framework in CAVs. Data selection reduces computational demand and accelerates training by prioritizing informative and safety-critical samples, while client selection broadens environmental representation and engages vehicles with sufficient computational and communication capacity to support global aggregation. The coordination of these processes enables balanced participation, comprehensive knowledge integration, and systematic utilization of available resources in the vehicular network. Beyond these core coordination mechanisms, communication and security/privacy impose system constraints that influence all stages of FL coordination in CAVs.

7) Communication

Communication plays a critical role in enabling FL across CAVs. Autonomous vehicles leverage cellular networks, including 5G and emerging 6G, as well as V2X infrastructures including DSRC, to exchange model parameters and aggregation updates while preserving data privacy [123], [124], [125]. FL introduces a specialized communication layer that operates independently of traditional vehicular messaging, focusing solely on transmitting model weights rather than raw

sensory data [126]. Therefore, communication must function reliably under real-world conditions of **mobility, variable latency, and constrained bandwidth**, ensuring that collaborative learning continues seamlessly without interfering with the vehicle's inference processes. Previous studies have explored FL communication under bandwidth and latency constrained networks, including strategies to **reduce staleness and improve synchronization efficiency** [55], [56], [127], [128], while QoS objectives have also been considered in related settings [129], [130]. In CAV deployments, the communication problem is coupled with resource constraints and safety-critical inference requirements. This motivates **system-level coordination** policies that jointly account for update staleness, network QoS, and on-board resource budgets.

Inference and training communications differ primarily in scope and timing. Training communication takes place periodically, involving the transmission of model updates between vehicles and a global or edge aggregator. These exchanges, though lightweight compared to raw data transfer, are highly dependent on stable network connectivity and efficient bandwidth utilization. The challenge lies in coordinating these updates under varying network conditions without compromising the continuous operation of the inference. Balancing these inference and training communication flows is essential to ensure consistent model improvement while preserving the reliability and responsiveness of the autonomous driving system.

The size of the model update determines the training communication. FLEdge reports per client model update can range from 0.3 MB for CNN/LSTM workloads to 308 MB for FLAN-T5 for larger models [75]. Therefore, per round model updates for each participating vehicle can range from tens or even hundreds of megabytes, given multiple on-vehicle models across modalities. Communication in FL involves transmitting **compressed model updates** [131], [132], [133], but **update frequency** still shapes the traffic, and jointly choosing compression and frequency under vehicular resource and connectivity constraints remains an open research problem.

In vehicular deployments, wireless links can be affected by coverage gaps and fluctuating throughput [56]. Such **intermittent connectivity can delay aggregation, introduce model staleness, and slow convergence**. To mitigate these challenges, hierarchical aggregation architectures can be employed, where RSUs or edge servers perform intermediate model fusion before synchronizing with the global aggregator. This layered approach reduces communication latency, improves scalability, and allows vehicles to contribute updates even under constrained network conditions [134]. However, coordinating communication between vehicles, RSUs, and centralized fleet management systems introduces additional complexity, requiring adaptive scheduling and topology-aware synchronization. Designing communication strategies that maintain inference continuity while optimizing **hierarchical learning** remains critical to achieving robust and efficient federated training in CAVs.

8) *Security and Privacy*

FL offers a promising framework for CAVs by reducing communication overhead and eliminating the need to share raw data with centralized servers. However, as CAVs collaboratively train models across **distributed and resource limited clients**, new security and privacy challenges emerge. The decentralized structure increases the vulnerability of the system to attacks during data acquisition, transmission of model update, and aggregation [135], [136], [137]. These risks highlight the need for robust defenses and reliable mechanisms to ensure safe and reliable deployment of FL in real-world vehicular networks.

The security dimension of federated learning extends across all phases of the collaborative training pipeline. In the data stage, manipulated or falsified sensor input can compromise the quality and consistency of local datasets [138]. During local training, adversarial participants can inject poisoned samples or backdoor patterns that degrade the accuracy of the model and propagate errors to the global model. In the communication stage, intercepted or modified updates can expose sensitive information or distort parameter exchanges. The aggregation process is also susceptible to collusion, where compromised clients bias the global model or infer information about others [139], [140]. Together, these vulnerabilities threaten the **integrity, confidentiality, and resilience** of the entire learning system. Although secure aggregation, encryption, and differential privacy can mitigate some of these risks, their computational cost and latency remain prohibitive for real-time vehicular applications.

Federated learning preserves privacy by keeping raw data in the vehicle. However, privacy risks in FL arise from the exposure of intermediate model updates that can be exploited to **infer sensitive information** such as driver identities, location trajectories, or visual details of surrounding environments. **Adversarial inference attacks** use these updates to reconstruct local data distributions or extract identifiable features from clients. Mitigation techniques including differential privacy, gradient perturbation, and secure multiparty computation reduce the likelihood of leakage, but may compromise model accuracy and increase computational complexity, which conflicts with real-time vehicular constraints [141], [142]. Emerging threats such as membership inference and property inference remain insufficiently addressed in current frameworks. These vulnerabilities emphasize the need for lightweight and adaptive privacy preserving solutions that align with the latency, energy, and scalability requirements of vehicular networks.

IV. TOWARDS DEPLOYABLE FL IN CAVS

The challenges and opportunities highlighted in Sections III-A and III-B provide a guiding framework for deployment-oriented FL in CAVs. Integrating FL into CAV systems has received comparatively less attention than FL framework strategies such as data/client selection, aggregation, and communication. Explicit coordination between real-time inference and on-vehicle training remains limited in both system and workflow level designs. The discussion highlights coordination mechanisms that make inference constraints explicit and account for on-vehicle resource budgets, storage and

staging costs, and network service quality, enabling federated training and model updating to operate alongside safety-critical inference. We present future directions in a priority order guided by deployment integration requirements.

To prioritize progress toward practical deployment, the foremost requirement is integrating federated training into the CAV stack without compromising safety-critical operation. To enable integration, software level mechanisms must be addressed first, as they are fundamental for ensuring safety and real-time guarantees. The integration at the software stack level requires RTOS and middleware support for training–inference coexistence, including (i) resource scheduling and allocation that bounds training interference on inference execution, (ii) bounded synchronization and data movement overheads across the runtime pipeline, and (iii) QoS-aware middleware that controls serialization, routing, and bandwidth so model update pipelines do not interfere with real-time dataflow. To validate that these mechanisms preserve real-time operation under on-vehicle training, deployment readiness should be evaluated under diverse workload intensities using metrics that quantify training–inference interference and real-time latency, such as end-to-end inference latency (p95/p99), jitter, and timing violations (deadline miss rate or worst case response time). Addressing these mechanisms is the first priority to realize stable, efficient, and safety-aware training–inference coexistence in CAVs.

In parallel with software stack integration, hardware resources, including computing and storage should ensure that on-vehicle training operates within bounded compute and energy limits under shared resources with inference. From the computing perspective, the immediate requirement is to make training execution predictable and controllable on heterogeneous on-vehicle compute resources that are shared with safety-critical inference. Therefore, future work should develop runtime compute coordination for training–inference coexistence that provides control over training placement across compute units and over training execution intensity, frequency, and duration so inference priority and timing guarantees are preserved. This requires designing dynamic training coordination mechanism that selects these training settings per vehicle and adapts them over time as workload demand and available compute and energy budgets change. Researchers should consider evaluating under diverse operating conditions that shift autonomy workload demand on heterogeneous hardware platforms, since both factors directly shape the compute and energy available for on-vehicle training. Under these operating conditions, evaluation metrics should be accelerator utilization, memory usage, training time variability, and power and thermal draw during training.

Storage is a critical deployment consideration for on-vehicle training. Future directions should develop storage management for limited capacity and endurance constraints, including near-term staging, retention, and compression that reduce storage demand while preserving training utility under sustained storage I/O load. To quantify storage overhead, evaluation should report storage residency time, cache and SSD pressure, staging backlog, and I/O intensity, and it should include endurance stress under representative workloads.

As federated training becomes an integral component of CAV systems, ensuring the security of data, models, and system behavior becomes a critical concern. In the absence of strong security and privacy mechanisms, the trustworthiness and safety of autonomous functions can be fundamentally compromised. Such an integration requires mechanisms that respect the resource constrained real-time demands of CAV platforms. Security and privacy support should therefore emphasize lightweight authentication and integrity protection for model updates, robust aggregation against poisoned or unreliable clients, and runtime safeguards that detect anomalous update behavior without violating real-time constraints. Their effectiveness should be evaluated by quantifying their latency and resource overhead on embedded platforms, including computing and memory cost and communication overhead, and by measuring robustness under representative adversarial and fault conditions such as update tampering, poisoning attempts, and intermittent client reliability.

Beyond system integration and trustworthiness, communication is a key priority, as inefficient model transmission or delayed coordination across vehicles can hinder the responsiveness and scalability of training at the edge. Future directions should emphasize scheduling and coordination policies that tolerate intermittent links and bandwidth variation, and performance should be evaluated using update latency, staleness, bandwidth usage, and partial participation under intermittent connectivity.

With the integration, security, and communication mechanisms established, attention turns to the design of the federated learning layer. Given the considerable progress toward developing the federated learning layer, the focus now refines its practical deployment aspects. However, much of this work has relied on centralized datasets and evaluation metrics that fail to capture the non-IID, resource constrained, and safety-critical nature of CAV deployments. Therefore, the focus should be on developing realistic, decentralized datasets and benchmarks that reflect non-IID distributions, sensor heterogeneity, and resource limitations inherent to CAV environments. These datasets are essential because re-evaluating existing aggregation strategies on realistic, decentralized datasets can clarify robustness under deployment heterogeneity and system-level constraints of CAV environments and guide necessary adaptations.

With deployment realistic benchmarks in place, both model and aggregation should be re-evaluated to ensure robustness under client heterogeneity and hierarchical coordination. The resulting insights should inform the data and client selection policies that enable scalable, fair and efficient learning across heterogeneous CAVs. However, the inherently heterogeneous nature of CAVs coupled with system and communication constraints introduces a fundamental scalability challenge for federated coordination. Addressing this scalability challenge requires asynchronous coordination mechanisms that remain stable under delayed and irregular client updates [143], [144]. Aggregation should therefore be assessed under delayed and irregular updates, with emphasis on update staleness, straggler driven participation variability, and convergence latency under realistic coordination constraints.

In hierarchical FL, aggregation is primarily constrained by two deployment factors. Local heterogeneity reflects differences across vehicles in sensing availability, data characteristics, and onboard processing, which can result in uneven update quality and tier dependent drift. Mobility driven participation variability reflects intermittent connectivity and vehicles dynamically entering and exiting RSU coverage, which introduces irregular participation and stale updates across levels [145]. Moreover, RSUs may function not only as intermediate aggregators but also as sensing nodes that provide complementary observations [146], further increasing cross-tier heterogeneity that hierarchical coordination must accommodate. therefore, hierarchical aggregation should be evaluated under multi-level coordination effects, including hierarchy-level participation variability and update staleness, to establish robustness under intermittent connectivity and heterogeneous resource budgets. When RSUs serve dual roles as both sensing nodes and intermediate aggregators, evaluation should also account for how RSU-collected observations and region-specific data availability affect tier-level update quality and aggregation bias. For hierarchical aggregation under intermittent connectivity, evaluation should present update staleness, straggler effects, and communication overhead along with accuracy, since these factors directly shape multi-level coordination behavior.

As we outlined the future research directions, we emphasize that deployment requires a systems foundation in which learning operates as an operational capability with constraints, coordination, and evidence of reliability and trustworthiness in real-world operation. Research practice should prioritize deployment-relevant design interfaces and evaluation criteria for coordination across the system stack and FL workflow. Progress should be judged against operational criteria that reflect real-world use rather than model-centric outcomes alone. The priorities outlined here provide a concrete basis for this standard and guide future work toward FL designs that are credible for real autonomy stacks.

V. CONCLUSION

Despite the strong and growing interest in autonomous driving, real-world adoption remains limited across regions, fleets, and operating conditions, making federated learning increasingly essential for scalable model improvement. This paper outlines the opportunities and challenges of enabling FL in CAVs from a deployment-oriented perspective across both CAV system integration and FL workflow design considerations. We present the CAV software-hardware stack with a federated coordination layer that supports federated training under practical resource and timing constraints. Our discussion highlights the constraints and design considerations that influence the feasibility of real-time, resource-aware FL in practical CAV operating conditions. Our work can guide future efforts on FL in CAVs toward solutions that account for the realities of deployment. Our findings have potential to be generalized to other cyber-physical systems, such as drones and mobile robots.

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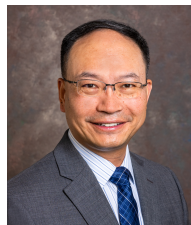
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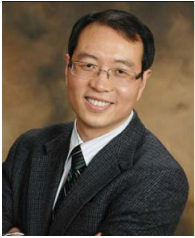
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